Amazon Elastic IP (EIP) — Important Interview Questions & Answers

1. What is an Elastic IP?
Elastic IP (EIP) is a static public IPv4 address provided by AWS. It can be associated with an EC2 instance and remains the same even when the instance is stopped and started.
Simple ga gurthupettuko:
Elastic IP ante AWS provide chese **permanent/static public IPv4 address**.

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2. Why do we use Elastic IP?
We use Elastic IP when an application needs a fixed public IP address. This is useful when users or external systems need to connect to the same IP continuously.
Simple ga gurthupettuko:
Application ki **same public IP always kavali ante EIP use chestham**.

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3. What is the difference between Elastic IP and Public IP?
A normal public IP can change when an EC2 instance is stopped and started. An Elastic IP is static and remains associated with our AWS account until we release it.
Simple ga gurthupettuko:
Normal Public IP change avvachu, but **Elastic IP fixed ga untundi**.

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4. How do you associate an Elastic IP with EC2?
First, we allocate an Elastic IP from AWS and then associate it with an EC2 instance or network interface.
Simple ga gurthupettuko:
First AWS nundi **EIP teesukoni**, tarvata **EC2 ki attach chestham**.

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5. Can we move an Elastic IP from one EC2 instance to another?
Yes. We can disassociate the Elastic IP from one EC2 instance and associate it with another EC2 instance in the same Region.
Simple ga gurthupettuko:
Oka EC2 nundi **EIP remove chesi**, another EC2 ki **attach cheyyachu**.

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6. What happens if an EC2 instance fails?
If the EC2 instance fails, we can associate its Elastic IP with a healthy replacement instance. Users can then continue using the same public IP.
Simple ga gurthupettuko:
Old EC2 fail ayina, **same EIP ni new EC2 ki attach cheyyachu**.

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7. What is Allocate Elastic IP?
Allocate means requesting a new Elastic IP address from AWS for our account.
Simple ga gurthupettuko:
Allocate ante AWS nundi **kotha Elastic IP teesukovadam**.

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8. What is Associate Elastic IP?
Associate means attaching an allocated Elastic IP to a supported resource such as an EC2 instance or network interface.
Simple ga gurthupettuko:
Associate ante **EIP ni EC2 ki attach cheyyadam**.

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9. What is Disassociate Elastic IP?
Disassociate means removing the Elastic IP association from an EC2 instance or network interface. The EIP itself is not deleted.
Simple ga gurthupettuko:
Disassociate ante **EIP ni EC2 nundi remove cheyyadam**, EIP ni delete cheyyadam kaadu.

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10. What is Release Elastic IP?
Release means returning the Elastic IP address to AWS. After releasing it, we can no longer use that EIP from our account.
Simple ga gurthupettuko:
Release ante **EIP ni AWS ki return cheyyadam**.

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11. Does Elastic IP cost money?
Yes. AWS charges for public IPv4 addresses according to its current pricing. Therefore, unnecessary Elastic IPs should be released.
Simple ga gurthupettuko:
Unused EIP ni AWS account lo unnecessary ga maintain cheyyakunda **release cheyyali**.

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12. Can one Elastic IP be associated with multiple EC2 instances?
No. An Elastic IP cannot normally be associated with multiple EC2 instances at the same time. It can be moved from one instance to another when required.
Simple ga gurthupettuko:
Oka EIP ni **multiple EC2 instances ki same time lo attach cheyyalem**.

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13. Is Elastic IP Region-specific?
Yes. An Elastic IP belongs to a specific AWS Region. It can be moved between supported resources within that Region.
Simple ga gurthupettuko:
EIP **oka specific AWS Region ki belong avutundi**.

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14. What happens when we stop and start an EC2 with Elastic IP?
The Elastic IP remains associated with the EC2 instance, so the public IPv4 address remains the same.
Simple ga gurthupettuko:
EIP unna EC2 ni stop/start chesina **public IP same ga untundi**.

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15. What happens when we stop and start EC2 without Elastic IP?
When an EC2 instance without an Elastic IP is stopped and started, its normal public IPv4 address can change.
Simple ga gurthupettuko:
EIP lekunte EC2 stop/start tarvata **public IP change avvachu**.

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16. Can Elastic IP be associated with a Network Interface?
Yes. An Elastic IP can be associated with an EC2 network interface and its private IPv4 address.
Simple ga gurthupettuko:
EIP ni **EC2 Network Interface ki kuda associate cheyyachu**.

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17. Should we use Elastic IP for every EC2?
No. We should use Elastic IP only when the application actually requires a fixed public IPv4 address. For many applications, a Load Balancer or DNS-based solution may be better.
Simple ga gurthupettuko:
Prathi EC2 ki EIP avasaram ledu; **fixed public IP requirement unte matrame use cheyyali**.

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18. What is the difference between Elastic IP and Route 53?
Elastic IP provides a static public IPv4 address. Route 53 is a DNS service that maps domain names to AWS resources.
Simple ga gurthupettuko:
EIP **IP address ni provide chestundi**, Route 53 **domain name ni resource ki map chestundi**.

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19. What is the difference between Elastic IP and Load Balancer?
Elastic IP provides a fixed public IP address, while a Load Balancer distributes incoming traffic across multiple backend resources.
Simple ga gurthupettuko:
EIP **fixed IP kosam**, Load Balancer **traffic distribute cheyyadaniki**.

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20. What is the difference between Elastic IP and Private IP?
A Private IP is used for communication inside the VPC. An Elastic IP is a public IPv4 address used for internet-facing connectivity.
Simple ga gurthupettuko:
Private IP **VPC internal communication kosam**, EIP **internet communication kosam**.

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21. Can Elastic IP communicate with the internet?
Yes. Elastic IP is a public IPv4 address, so it can be used for internet-facing communication when the network and security configuration allows it.
Simple ga gurthupettuko:
EIP anedi **internet-facing public IP address**.

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22. Can we release an Elastic IP and get the same IP again?
No. After releasing an Elastic IP, AWS may assign that address to another customer. We should not expect to get the same IP again.
Simple ga gurthupettuko:
EIP release chesina tarvata **same IP malli vastundi ani guarantee ledu**.

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23. What happens if an Elastic IP is not associated with a resource?
An unassociated Elastic IP remains allocated to the AWS account and may incur public IPv4 charges. If it is not required, we should release it.
Simple ga gurthupettuko:
Use cheyyani EIP unte **unnecessary charges avoid cheyyadaniki release cheyyali**.

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24. How would you handle an EC2 server failure using Elastic IP?
I would launch or identify a healthy replacement EC2 instance and associate the existing Elastic IP with it. This allows clients to continue using the same public IP.
Simple ga gurthupettuko:
EC2 fail ayite **same EIP ni healthy replacement EC2 ki attach chestham**.

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25. What would you do if the EC2 public IP keeps changing?
If the application requires a fixed public IPv4 address, I would allocate an Elastic IP and associate it with the EC2 instance.
Simple ga gurthupettuko:
Public IP frequently change avute **fixed IP kosam EIP use chestham**.

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26. What would you do with an unused Elastic IP?
If an Elastic IP is no longer required, I would release it from the AWS account to avoid unnecessary public IPv4 charges.
Simple ga gurthupettuko:
Unused EIP ni AWS account lo maintain cheyyakunda **release cheyyali**.

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27. What is the most important thing to remember about Elastic IP?
Elastic IP is a static public IPv4 address that can be associated with supported AWS resources. It is mainly used when an application needs a fixed public IP address.
Simple ga gurthupettuko:
Elastic IP ante **application ki fixed public IPv4 address kavali appudu use chese AWS IP**.
ChatGPT is AI and can make mistakes.